“21 reasons why to chose NTEGRA AURA”

17th July 2008




NT-MDT
Molecular Devices and Tools for NanoTechnology

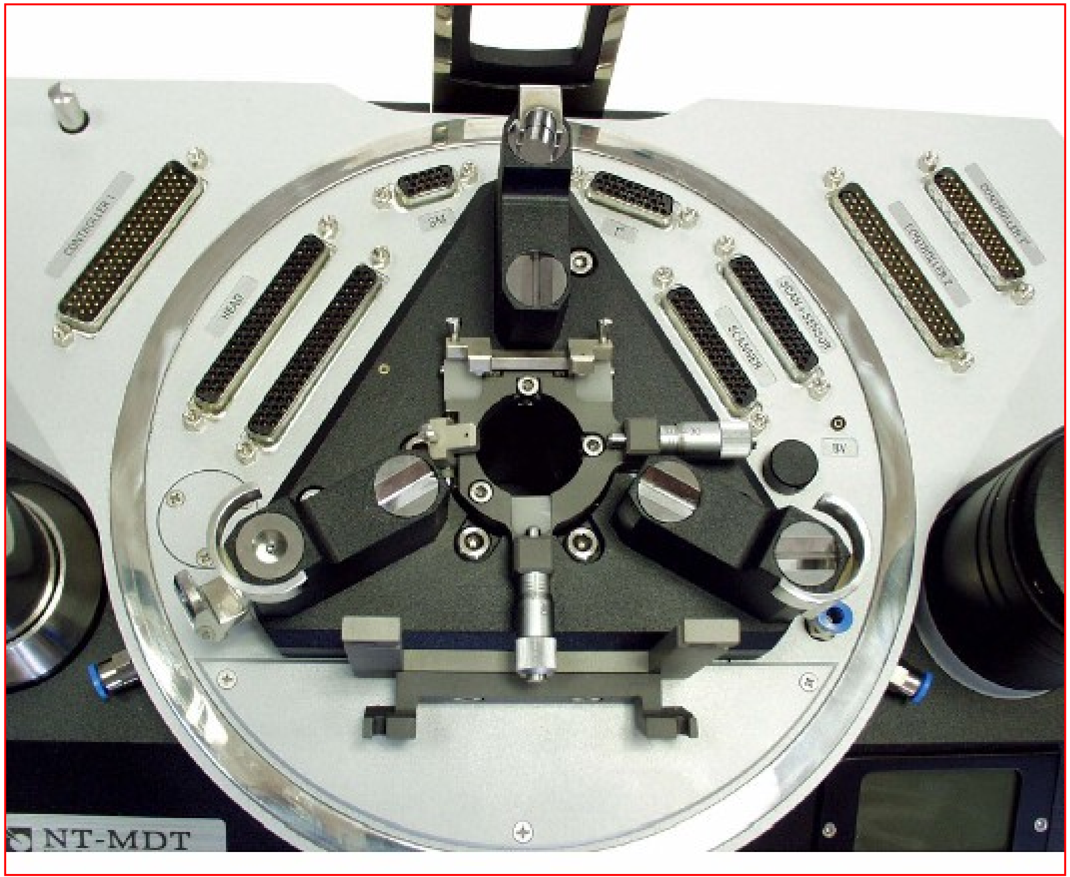
NTEGRA
PROBE NANOLABORATORY

Your choice is granted

1st reason why: - Design

Highly professional design
Ergonomic and esthetic design solutions make the daily work with the system easy, comfortable, and prestigious.

2nd reason why:- Materials



- Universal NTEGRA basement is symmetric. This feature occurs to be of extreme importance when <u>high stability</u> for ordinary measurements as well as at changing temperature is required.
- There is no preferable direction for drifts
- Another essential point is material the basement is made of. It is <u>titanium</u> that is perfectly balanced with piezoceramic in terms of thermal expansion.
- State of the art mechanics

A little more convenience…

- The system face panel is equipped with the LCD monitor that represents environmental conditions such as the temperature and humidity.
- The cover of LCD is scratch proved being made from sapphire glass.

3rd reason why:- Vacuum

Simultaneous high resolution optical access under vacuum.

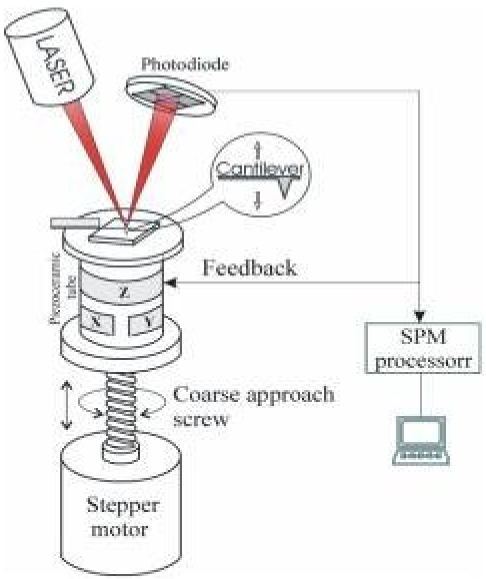
Special cast metal hood allows operation in a controlled gas atmosphere or in vacuum ($5x10^{-4}$ Torr).
It also serves as acoustical isolation and electrical shield

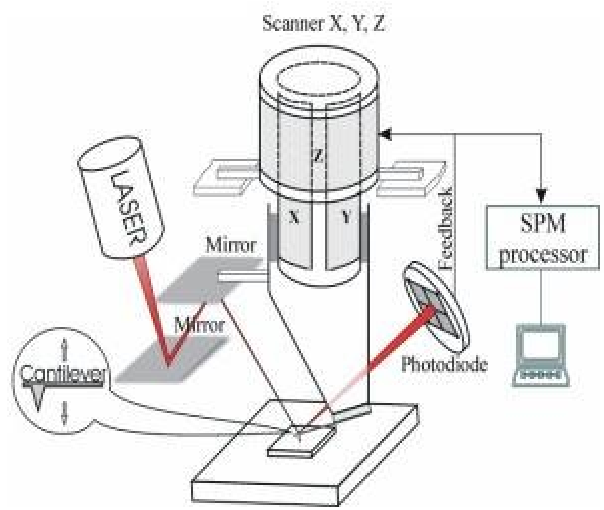
inlet/ outlet pipes enable to modify gas composition during the experiment, For Aura configuration these are not used, but there is KF 25 flange on a back of base.

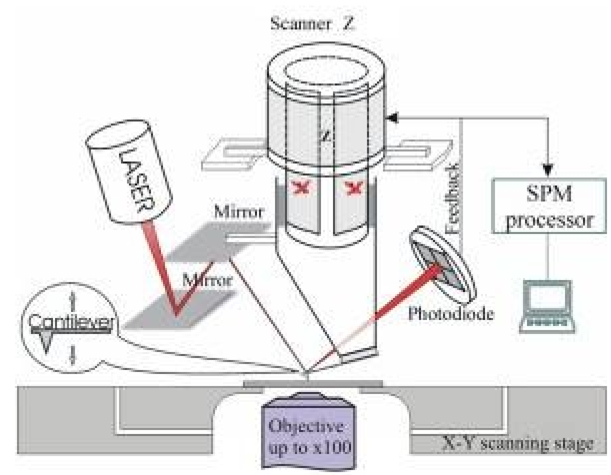
4th reason why: - Flexibility

Flexibility – 3 scanning principles included in 1 scanning microscope!

Standard included	Standard included	Optionally available







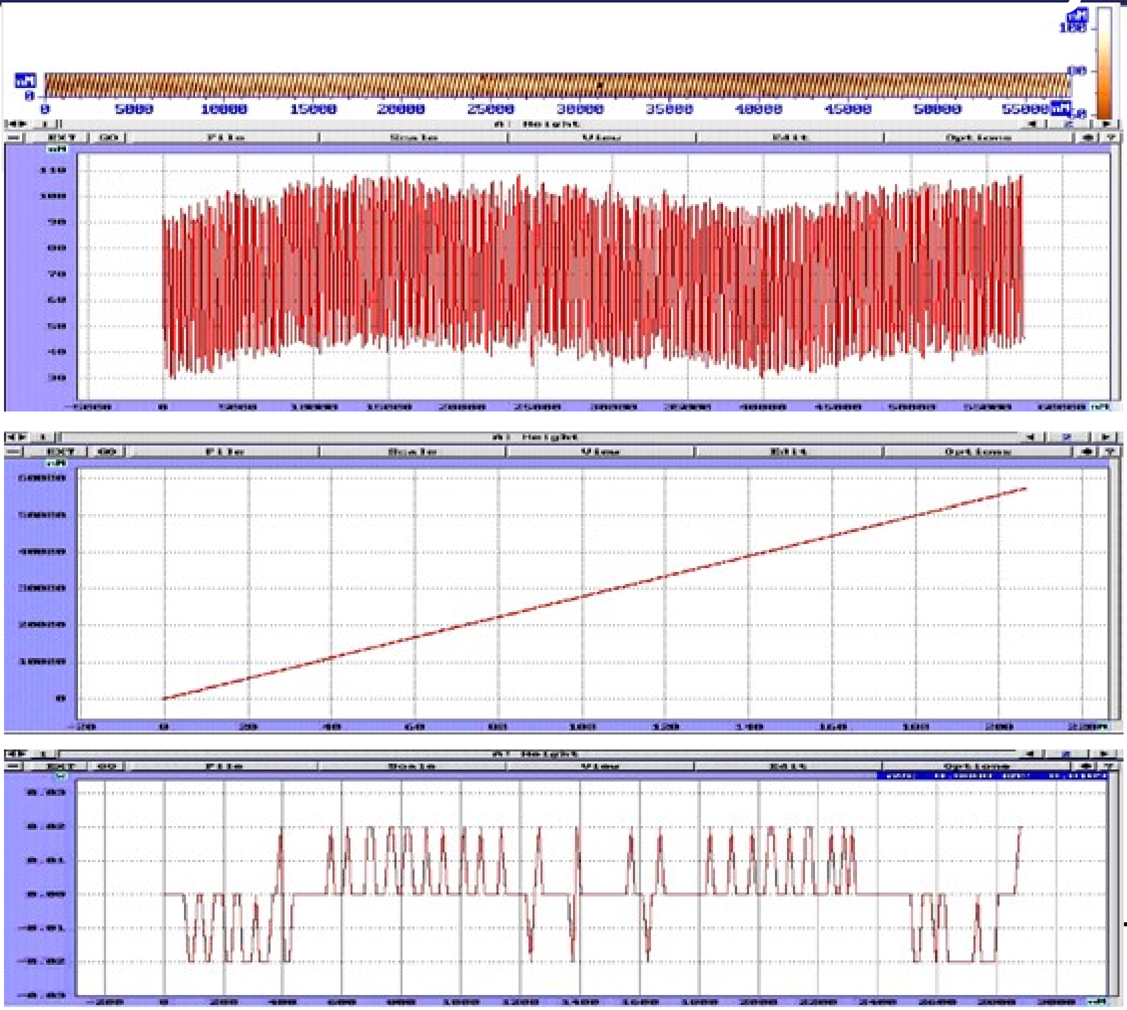
Scanning by the sample configuration. X-Y-Z tube on the bottom . Provides “stiff ” construction for Force-distance curves measurements	Scanning by a tip configuration. X-Y-Z piezotube on the top – Provides flexibility in sample size, and usage of liquid cells or heating/cooling stages.	Scanning by a sample, using flat X-Y scanning stage and only Z piezo from the top. Provides absolute no “cross talk”, and extrimely important for scanning large flat areas

4th reason why: - Flexibility

Flexibility – Does not mean Universality , it's true that there is nothing excellent if something is universal.
With Ntegra we are talking about completely different concept in scanning probe microscopy we are bringing on a market the platform technology that has in the core high resolution module and then with smart design we use the same platform in different microscopy fields without sacrificing initial performance .
With Ntegra we offer our expertise collected over 18 years of development in SPM field integrated in to powerful software with respect of user friendliness and easy to use for fast start up and obtaining results.

Thus we supply our customers with state-of-the-art proven equipment for laboratory excellence.

5th reason why: - World the best Scanner Accuracy



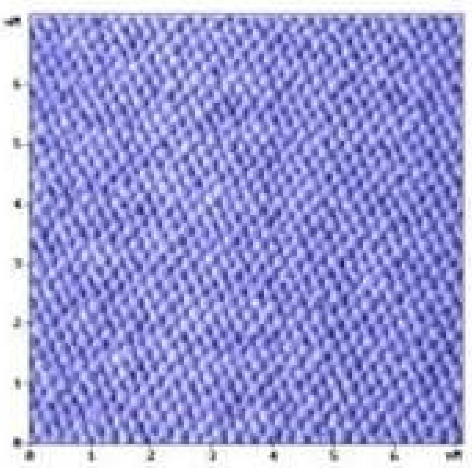
55 um Topography and Section of 256 nm pitch grating, 4000 points

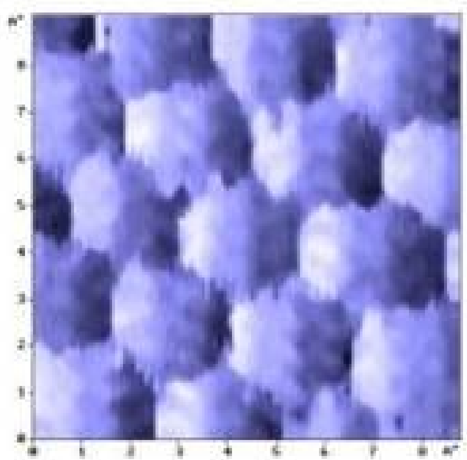
X coordinate peak position vs number (200 total)

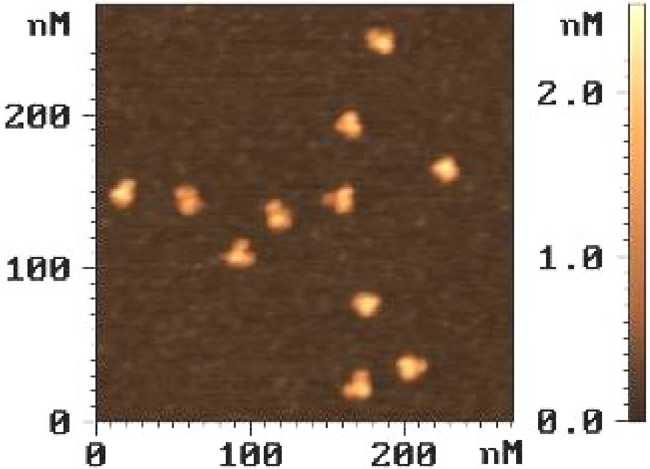
+/-0.05% XY residual nonlinearity

LOWEST noise capacitance sensors – 2 ppm i.e. 0.2nm for 100um scanner

6th reason why: - Highest Resolution







Atomic resolution on graphite (HOPG) LFM

Molecular resolution IgG proteins

The structures are just the same as revealed by X-ray crystal structure analysis

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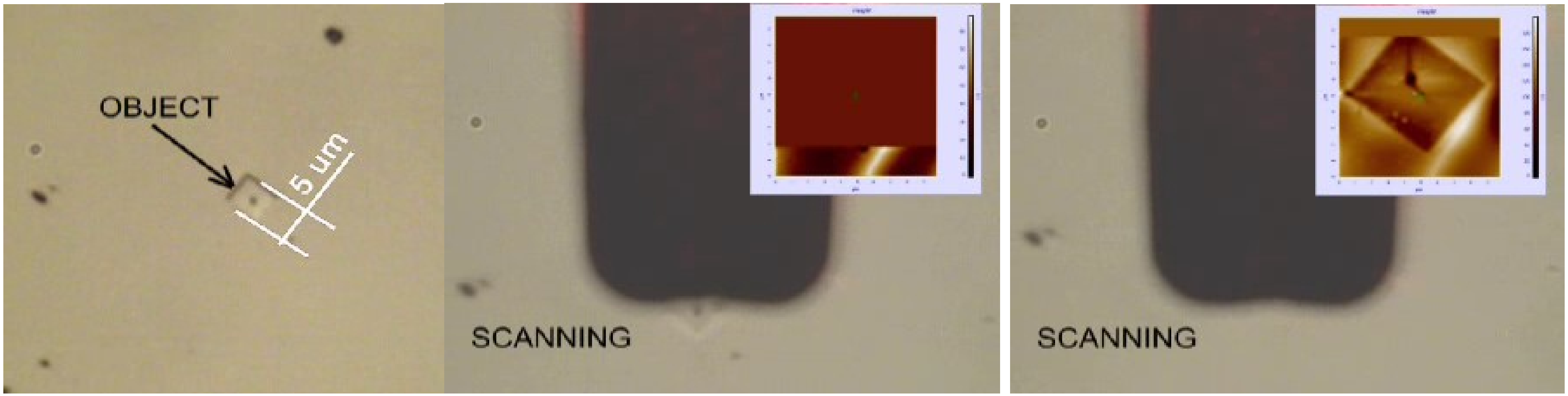
7th reason why: - Modern SPM Controller

- STM
- Contact AFM
- Lateral Force Microscopy
- ResonantMode -Semicontact
- Non-contact AFM mode
- Phase Imaging
- Force Modulation (viscoelastisity)
- Magnetic force Microcopy
- Electrostatic Force Microscopy
- Adhesion Force Imaging
- AFM nano-Lithography-Force and Voltage
- Scannng Spreading Resistance Imaging (SSRI) (10pA-5nA)
- Femtocurrent C-AFM (25fA-100pA) – (Optionally available)
- Scanning Capacitance Imaging (SCI)
- Scanning Kelvin probe microscopy (SKM)
- Force distance curves
- I/V spectroscopy, I(Z) spectroscopy etc.
- Force Volume
- Piezoresponce Mode
- Sample heating /cooling (-30+300 C) (can be included in delivering system)
- Electrochemistry (can be included in delivering system)
- SNOM (can be included in delivering system)
- Atomic force acoustic microscopy (AFAM) (can be included in delivering system)
- Scanning Thermal Microscopy (SThM) (optionally available)


Signal access module

SPM workstation BL222RNTF

8th reason why: - Perfect Optics



Optics for rough sample positioning is easy adjustable, there are 3 different upright microscopes available with Ntegra – standard with 3 and 1 micrometer resolution and optional 0.4 micrometer resolution which is the best available integration of high resolution optics with AFM.

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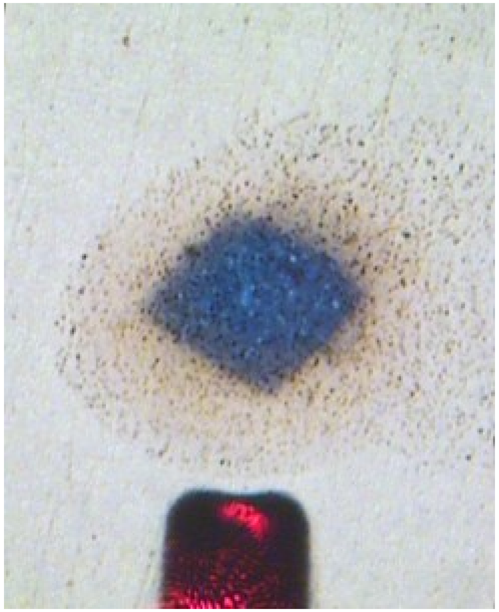
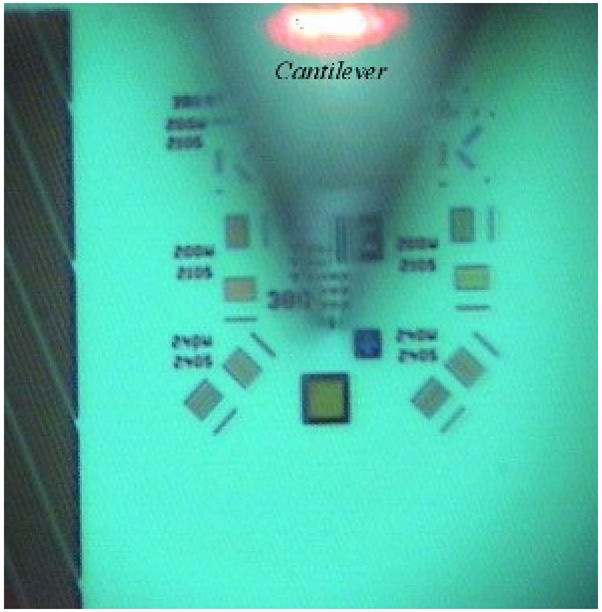
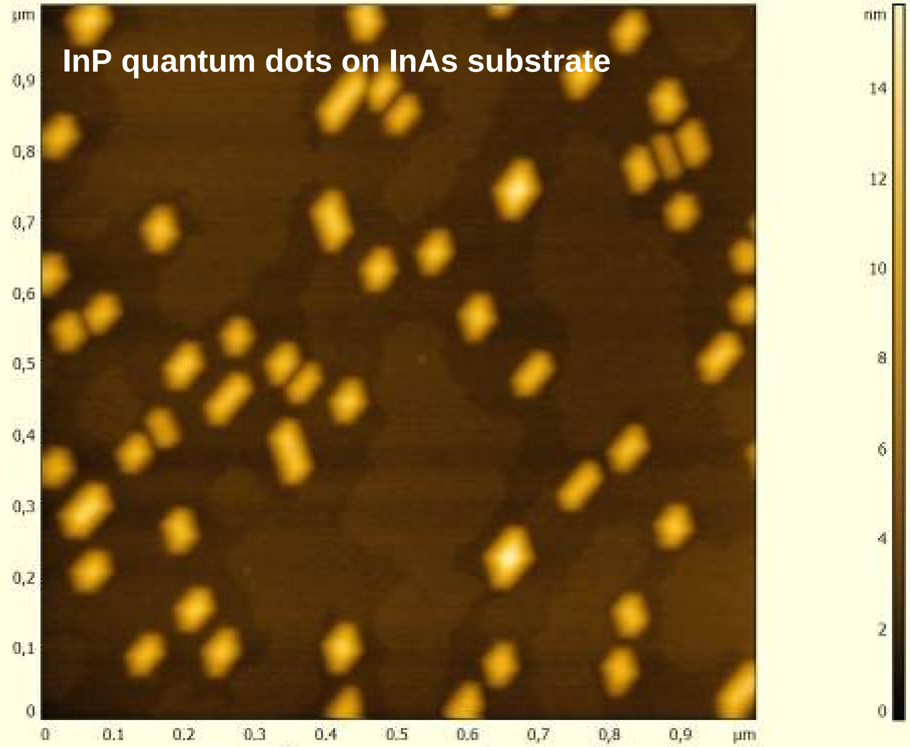
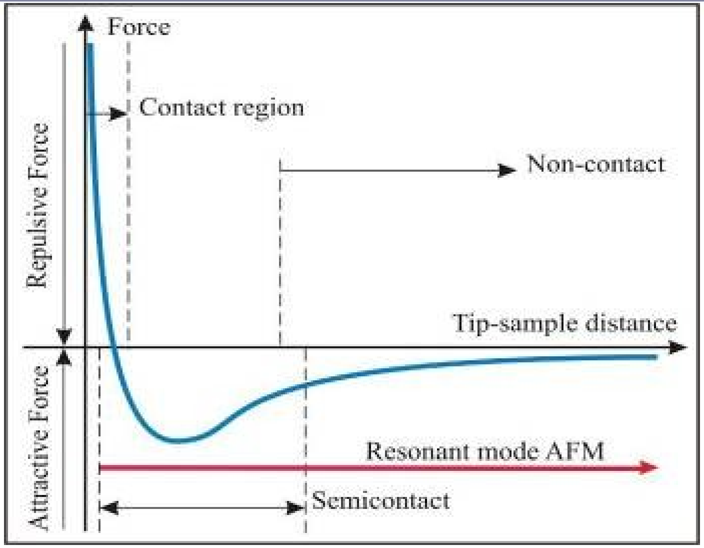


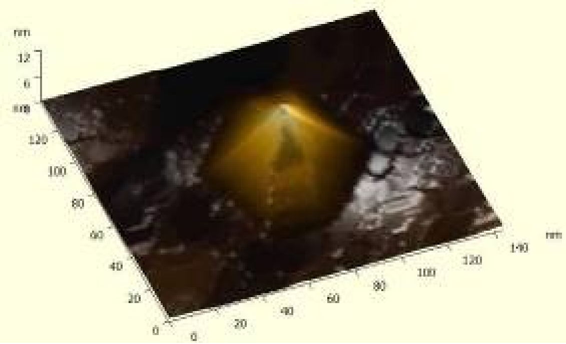


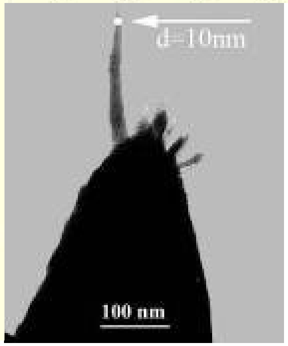
Optics for rough sample positioning is easy adjustable, there are 3 different upright microscopes available with Ntegra – standard with 3 and 1 micrometer resolution and optional 0.4 micrometer resolution which is **the best available integration of high resolution optics with AFM.**

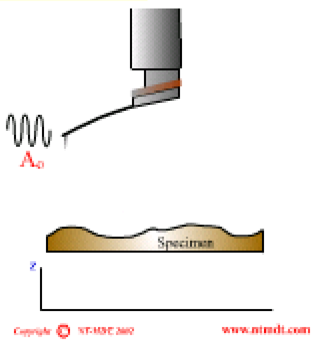
9th reason why: - Smart Approach







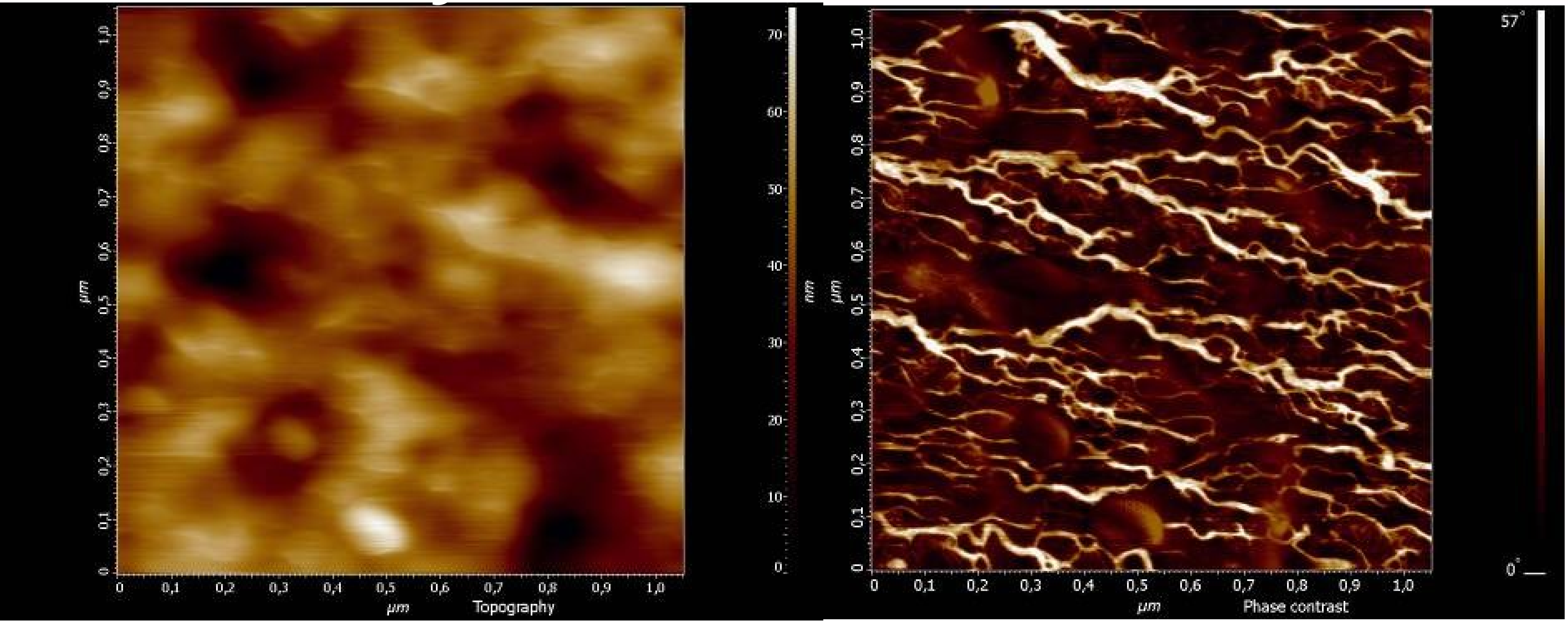




Lennard-Jones potential force between tip and sample

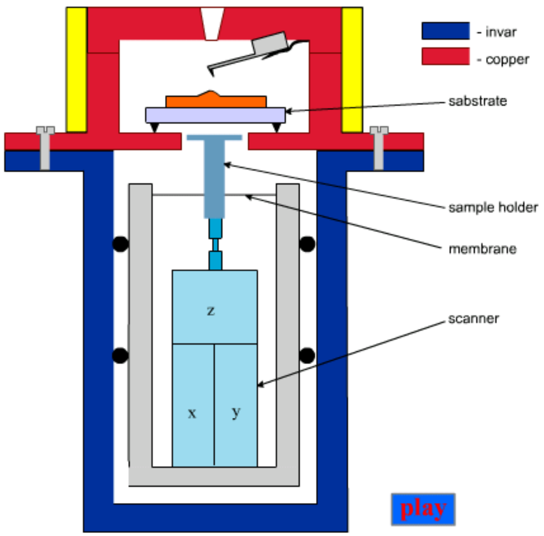
We do not damage even specialized extreme sharp tip during approach thus with NT-MDT equipment users are able to get stable and reproducible high resolution.

10th reason why:- Advanced Phase sensitivity



Block-copolymer ,Scanning area 1x1 micrometer
Cantilever force constant 5n/m
Driving voltage 0,08V, Amplitude reduction to free amplitude 20%
Phase sensitivity is 0.01 degree
Resolution on phase image 3-5nm of single lamellas.

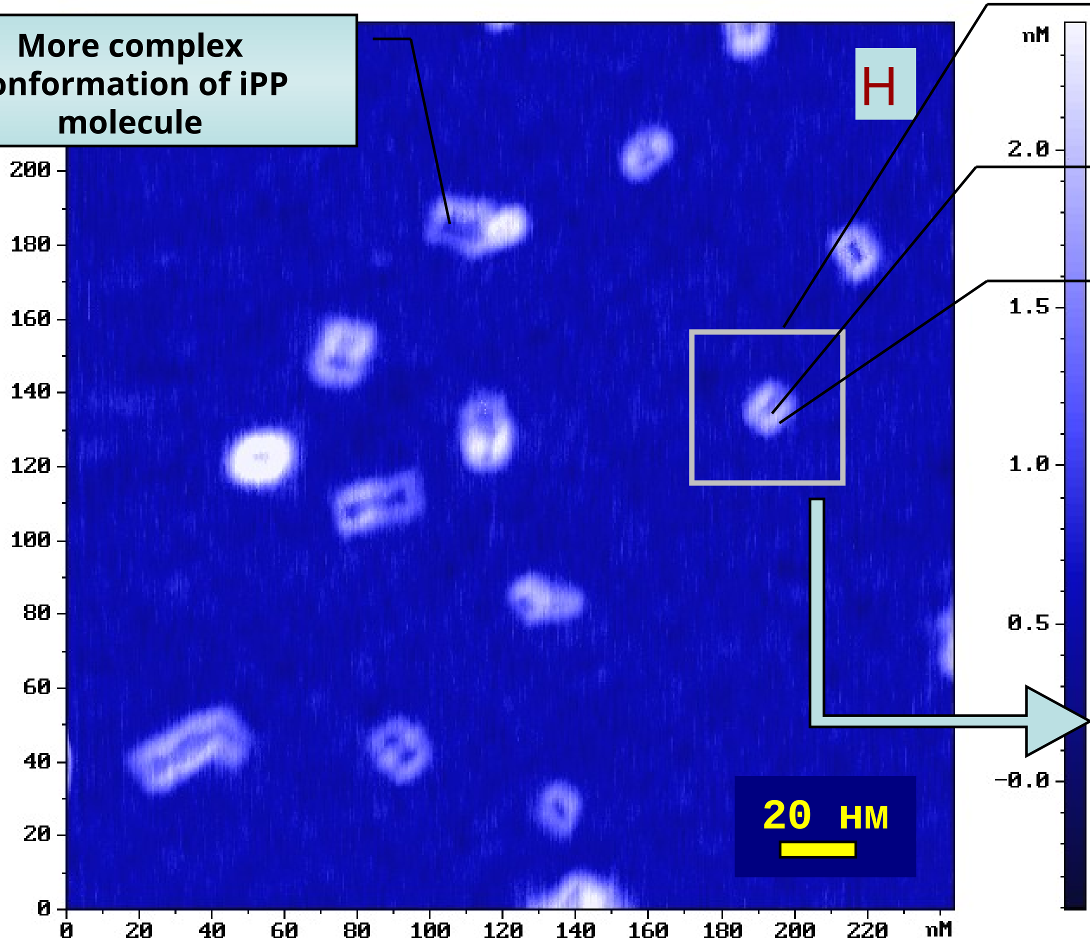
11th reason why: - Unique performance



Parameter	
Drift XY, less than	**3 nm/hour**
Drift Z, less than	1 **nm/hour**
Thermal drift XY, less than	**20 nm/°C** (with special adjustment about 10 nm/°C)
Thermal drift Z, less than	10 nm/°C

11th reason why: - Unique performance

More complex conformation of iPP molecule

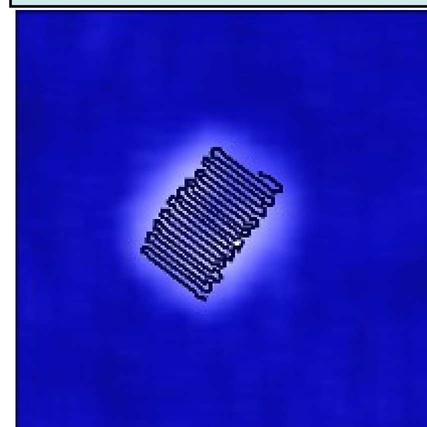


The “simple” crystalline conformation of single iPP molecule

The central (crystalline) part of the nanocrystallite

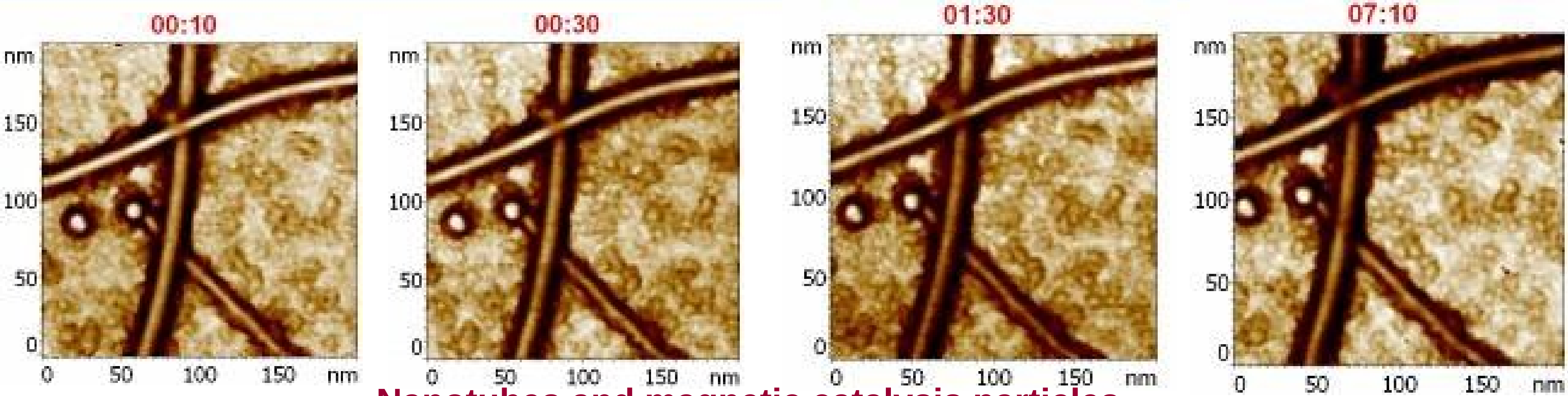
The “amprphous” part (the fold area)

The interpretation : single polymer chain, folded on indelf many times



AFM image of world highest resolution – single-molecule crystallites of isotactic polypropylene on mica

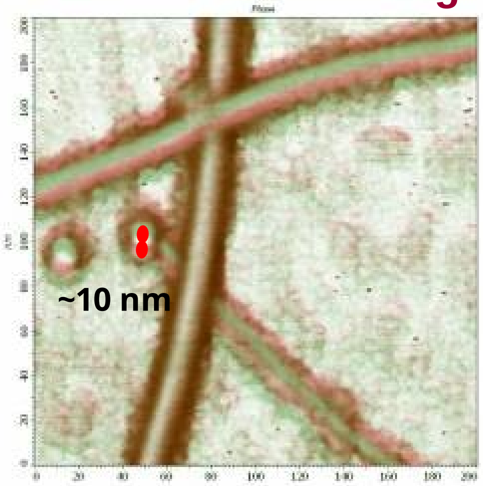
11th reason why: - Unique performance



Nanotubes and magnetic catalysis particles.
This system was monitored during 7 hours !

Room temperature

Scan size:
150x150 nm



Two scans compared (7 hours from each other).

Shift is <10 nm (2.4 nm per 100 min) !

University of California, Irvine, March 2006

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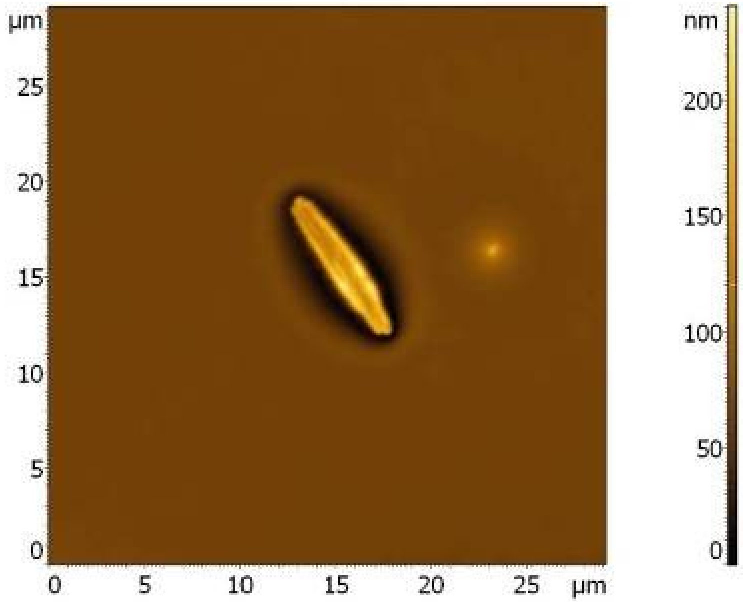
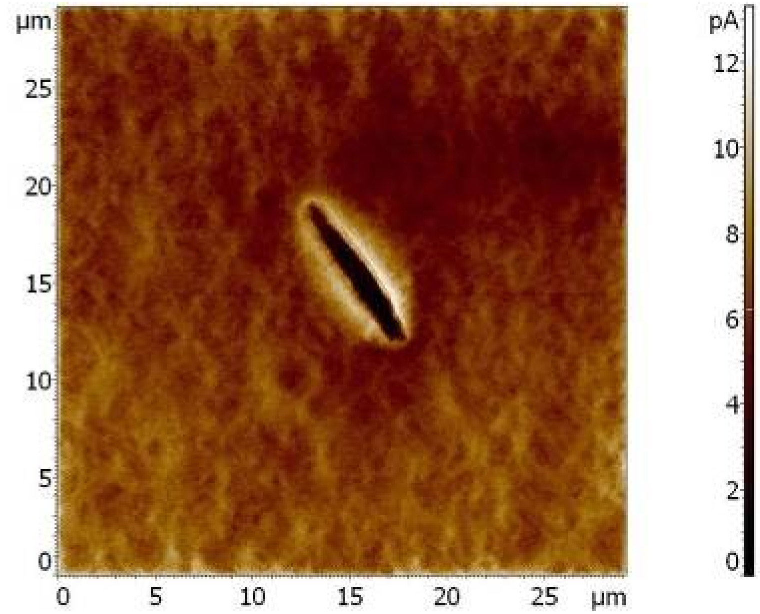
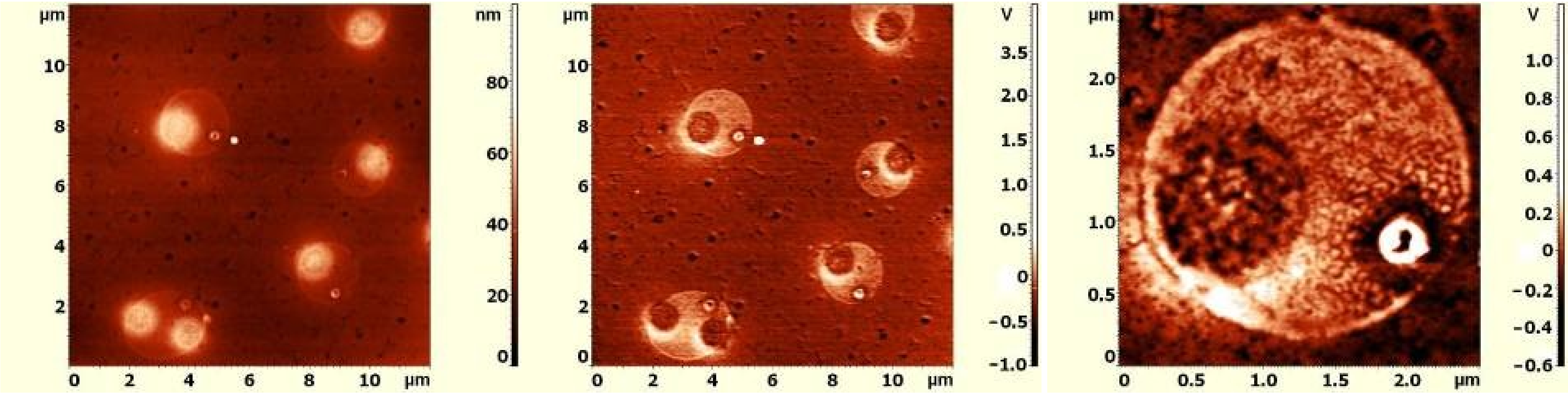




The best performance in Femto-current range of Conductive AFM
25fA in 100Hz bandwidth

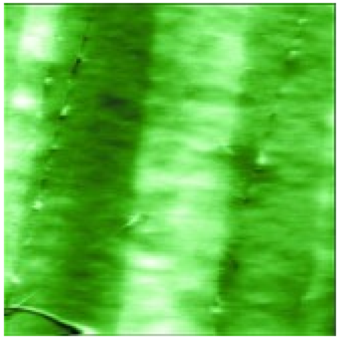
11th reason why: - Unique performance



Phase separated blends of Ru-PPy/PCBM (Ratio 1x1) semiconducor polymer. Topography and surface potential images, scan size 12x12 mm, Right image is surface potential image, zoomed area, scan size 2.5x2.5 mm, resolution better then 40 nm in Kelvin Probe microscopy on Polymers.

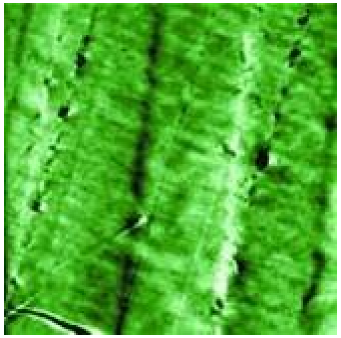
12th reason why: - AFAM

Atomic Force Accoustic Mocroscopy
Quantitative Messung mit Kontaktresonanzspektroskopie

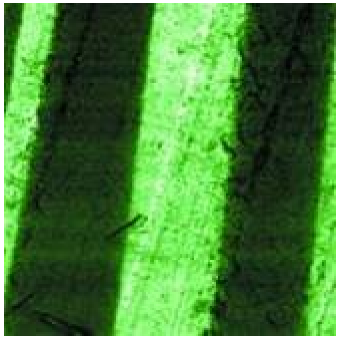


Topography

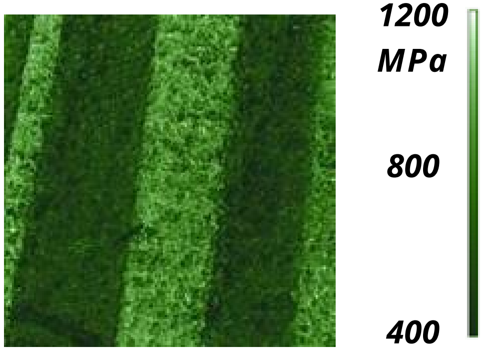
Phase imaging



Force modulation



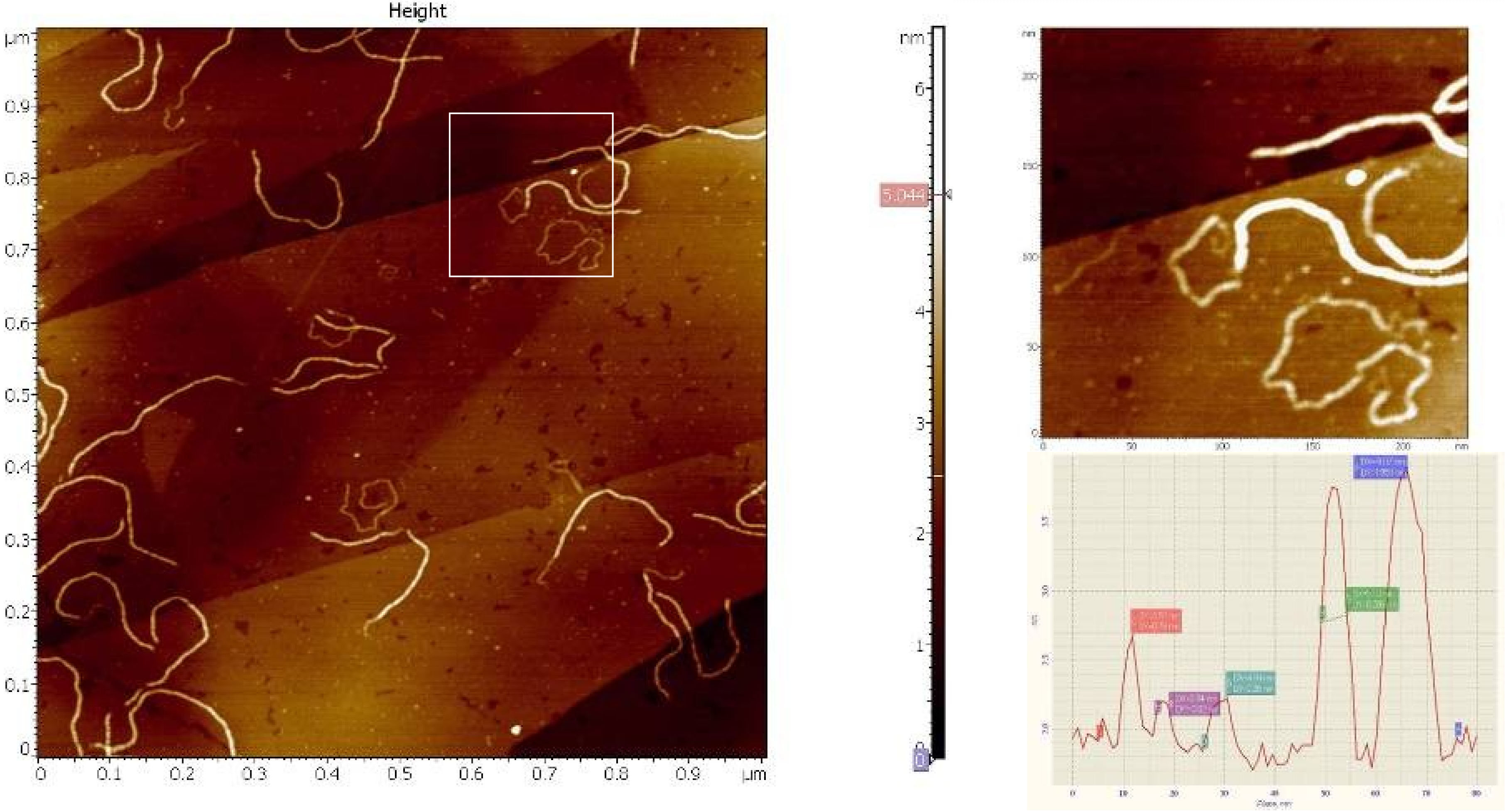
AFAM



Young modulus

Stripes of low and high density polyethylene with different elasticity. Scan size 47x47 um.
Stripes are well contrasted in AFAM but hardly visible in other methods

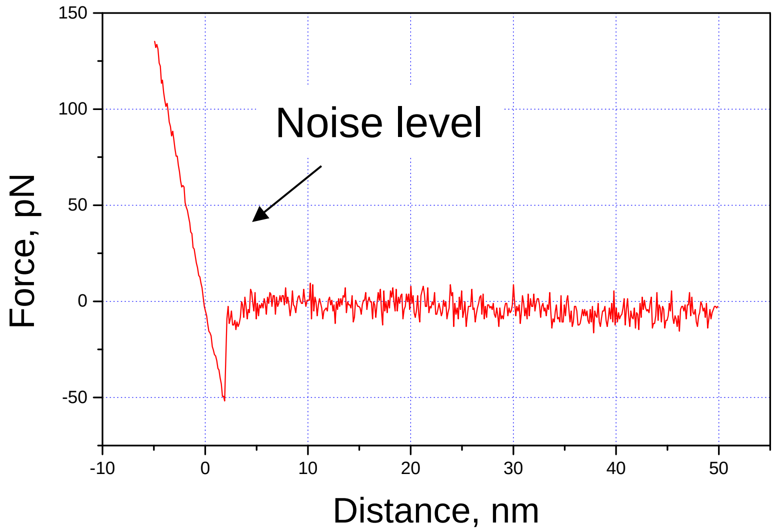
13th reason why: - One step Zoom



AFM image of DNA Triplex molecules deposited on HOPG.
Image quality 1024x1024 data points.

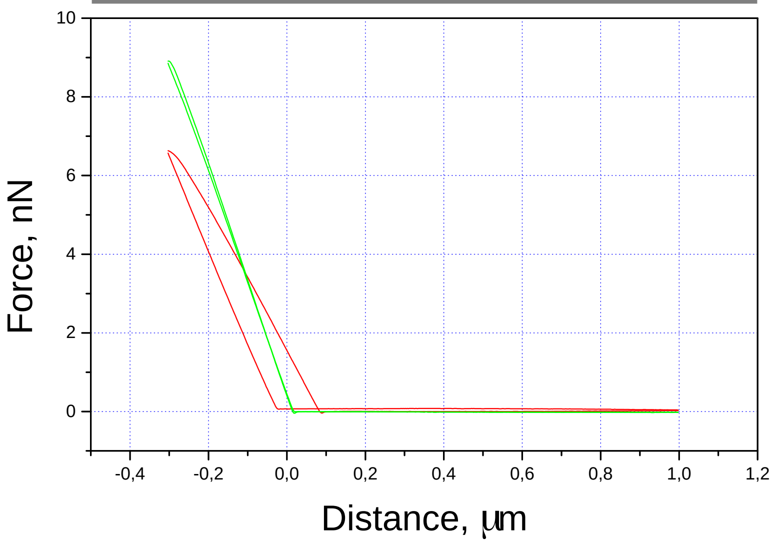
14th reason why: - Advanced Force spectroscopy

Sensitivity



Force-distance curve taken in liquid by silicon nitride triangular probe with 30 pN/nm force constant.
Here force resolution is limited at the level about several piconewton due the thermal noise of the probe. The RMS noise of force is smaller then 3pN.

Precision



Force-distance curves with close-loop on (green) and close-loop off (red).

With Ntegra we have scanner linearization in all 3 directions, for Z axis it is better then 1% residual non-linearity

Sensor noise < 0.06nm

15th reason why: - Scripting

Main Menu

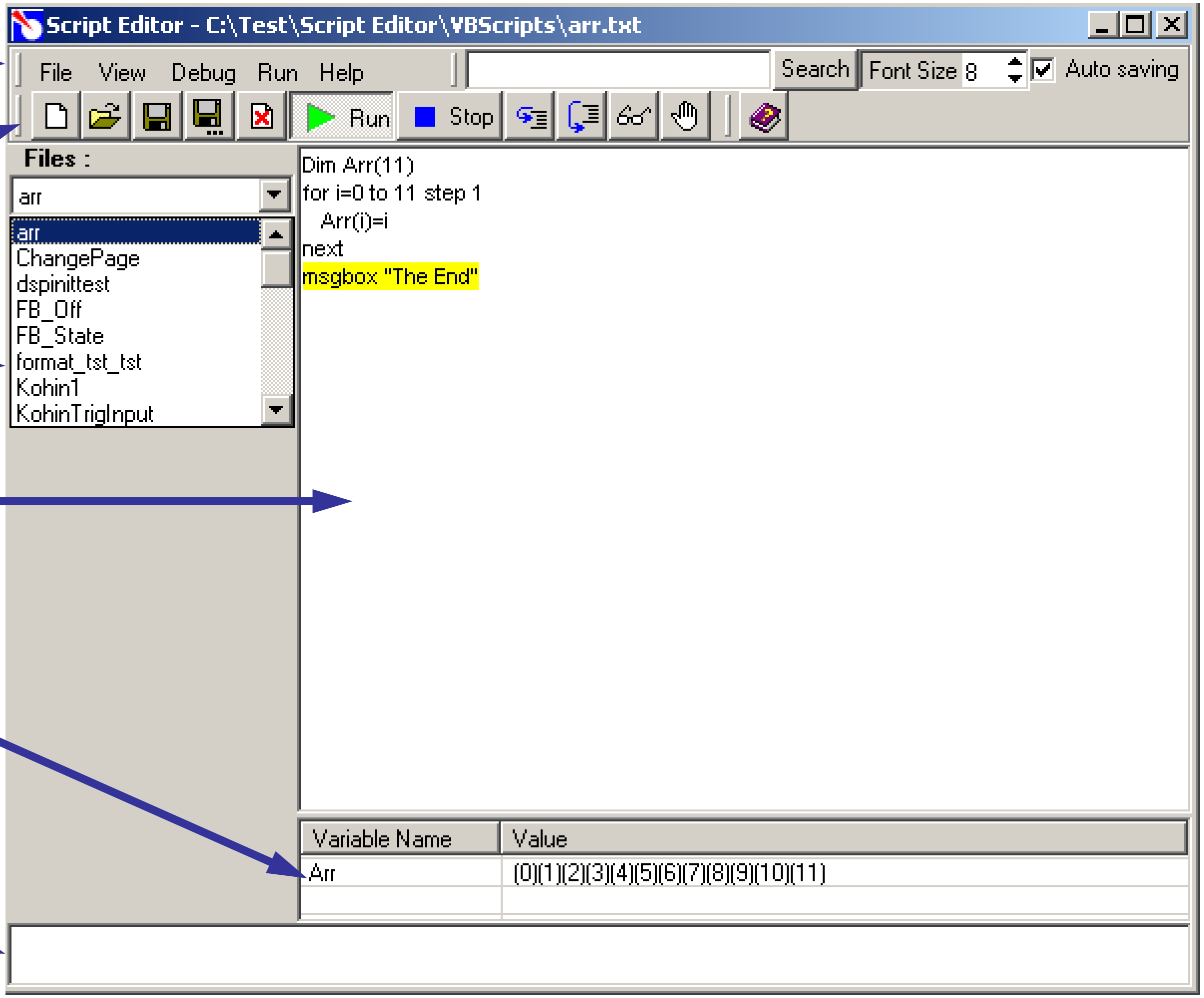
Operation Panel

List of Scripts

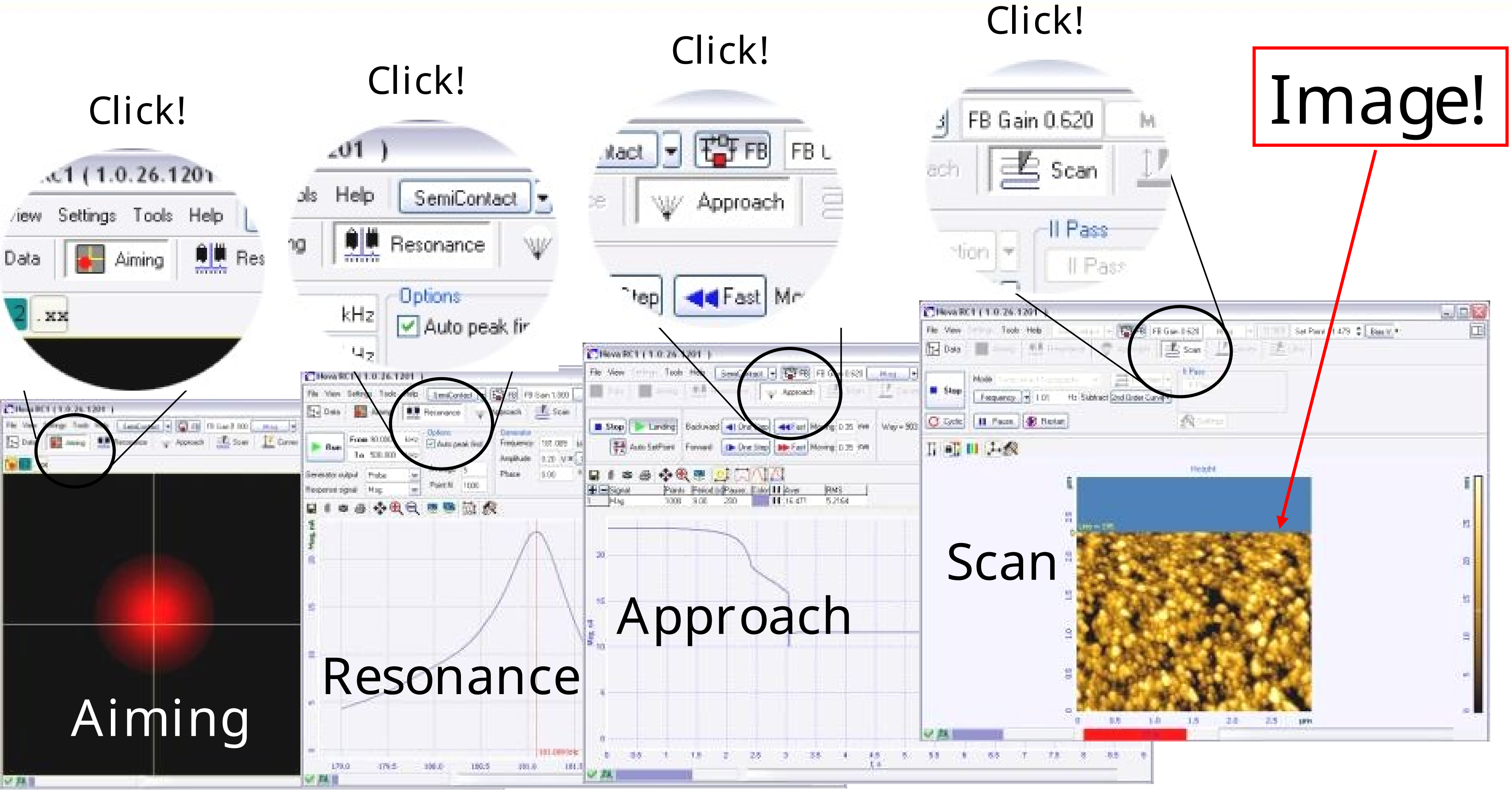
Editor

Variable Watch Window

Error Message Window



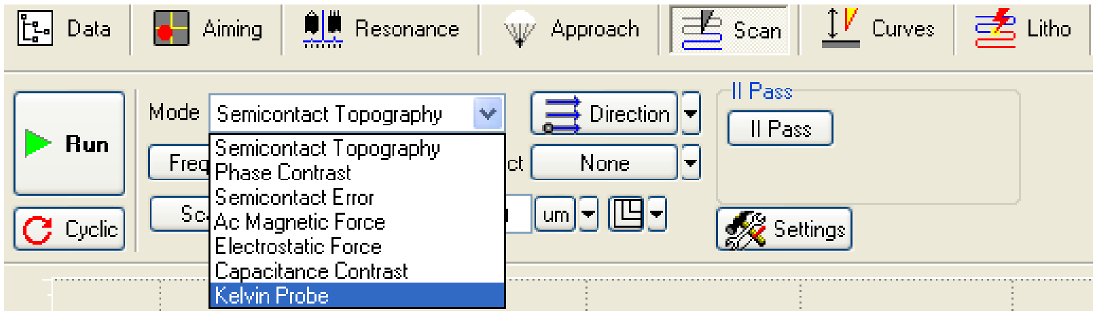
16th reason why: - Ease of use software



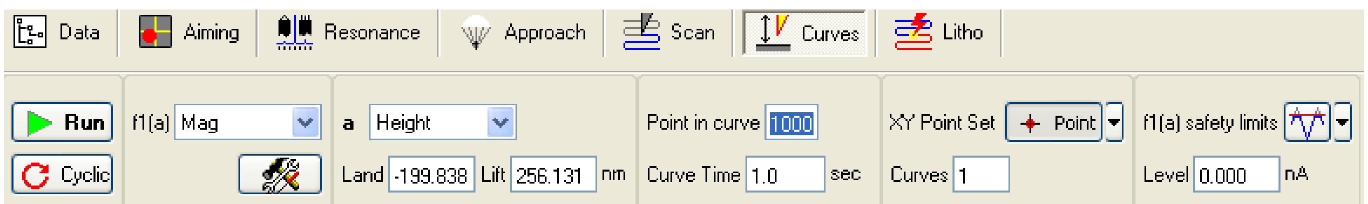
Tab-style windows configuration provides easy and fast step-by-step aim achievement

16th reason why: - Ease of use software

Even advanced modes are already programmed



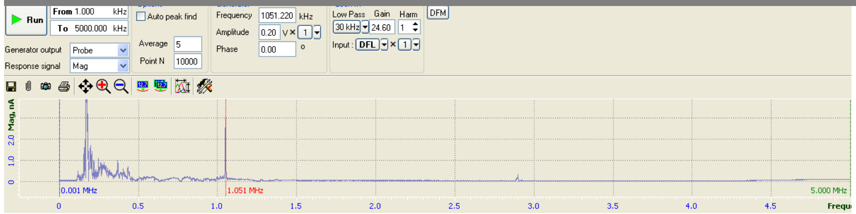
Force curves “Coarse ”and “Fine” modes included

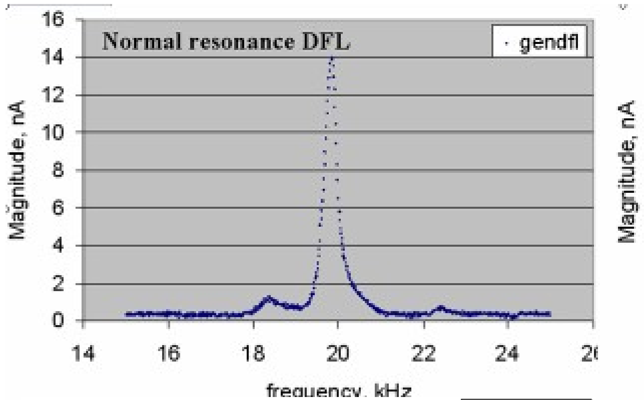


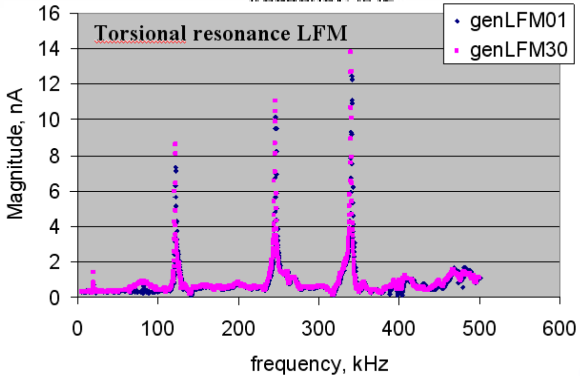
User can have all control for trace and retrace curve with velocity and delays in contact point.

17th reason why: - Flexibility in Software

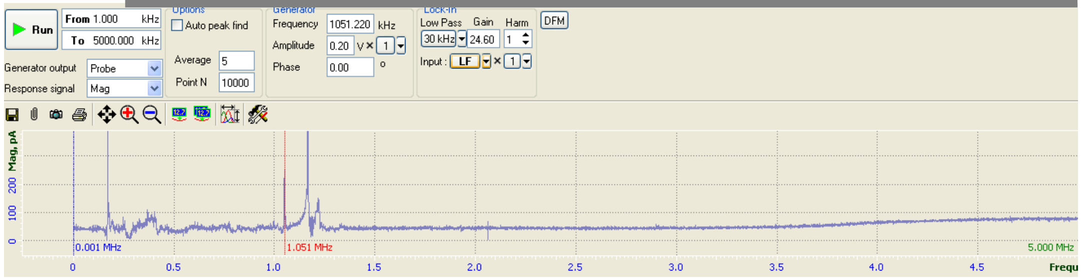
Frequency range up to 5Mz for normal oscillations



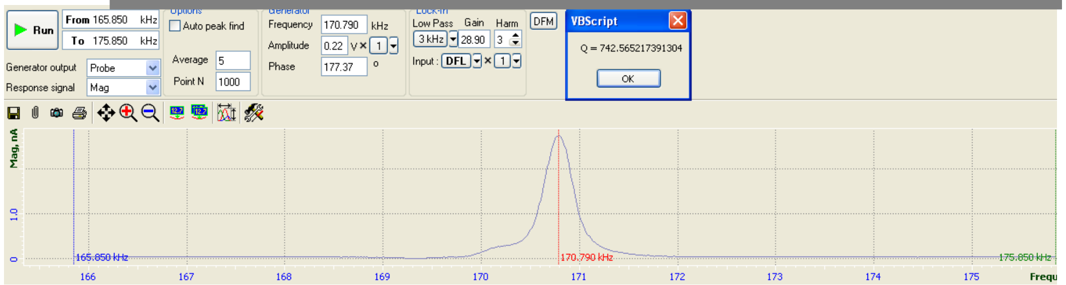




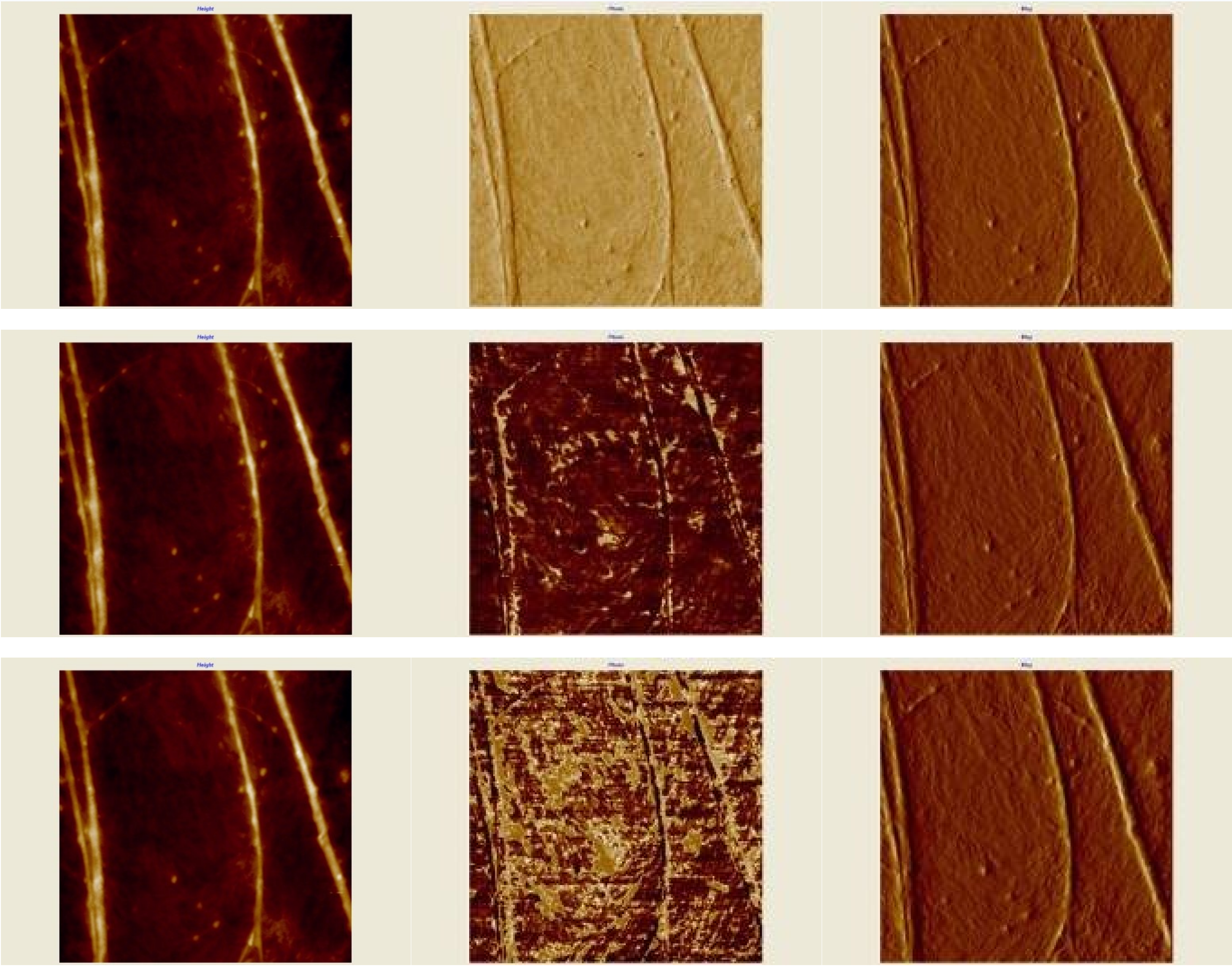
Frequency range up to 5Mz for lateral oscillations



Higher harmonics possibilities up to 9th



18^{th} reason why:- Imaging on higher Harmonics



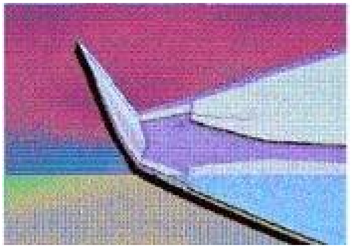
1st Harmonic

3th Harmonic

5th Harmonic

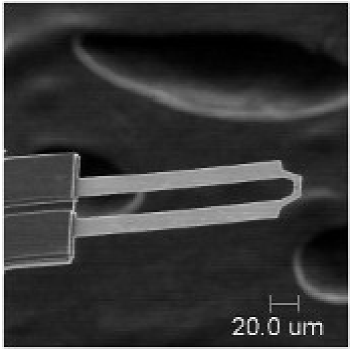
Stretched PE, scan size 5x5 micrometers, middle column is phase at different harmonics, used standard cantilever

19th reason why:- Integrated with SThM



Probe:

Max. Temperature:	160°C (dependent on probe)
Spring Constant:	~5 N/m
Resonant Frequency:	~50 kHz
Tip Radius:	<100 nm
Tip Height:	10 microns
Cantilever Length:	150 microns

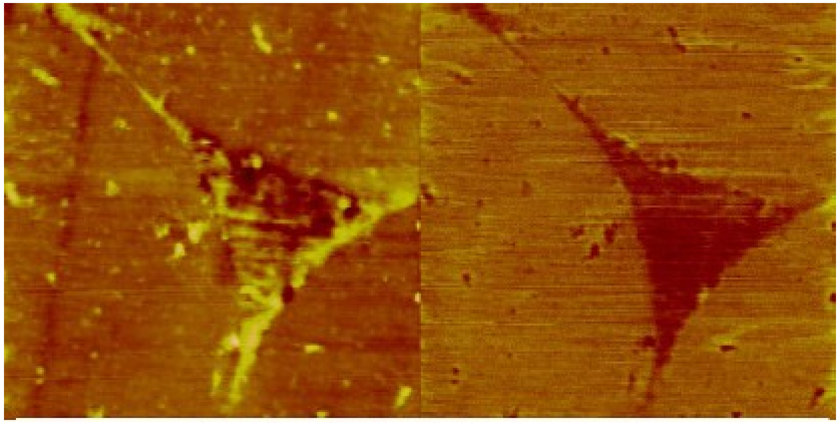


High temperature probe:

max. 450°C operating temperature

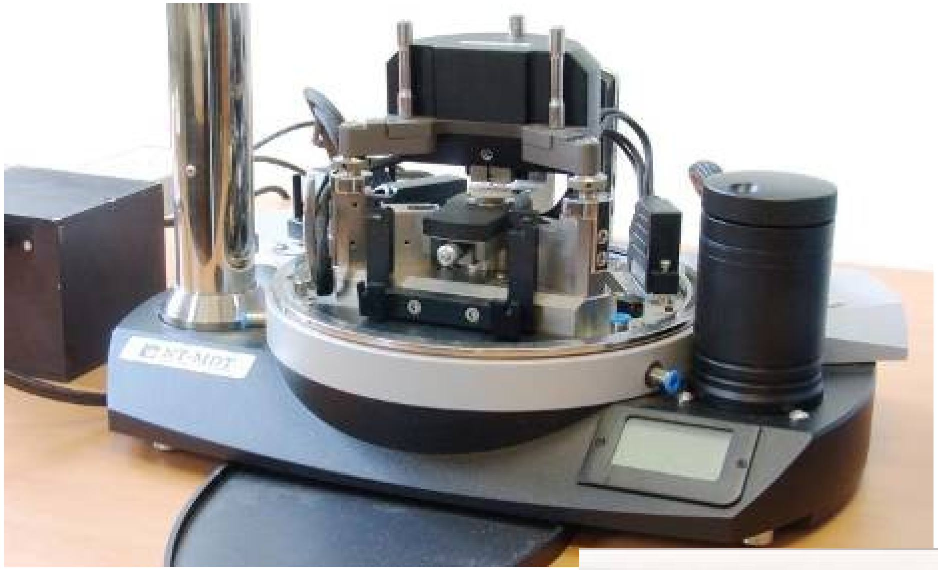
approx. 550°C for cleaning procedures

tip height approx. 1µm

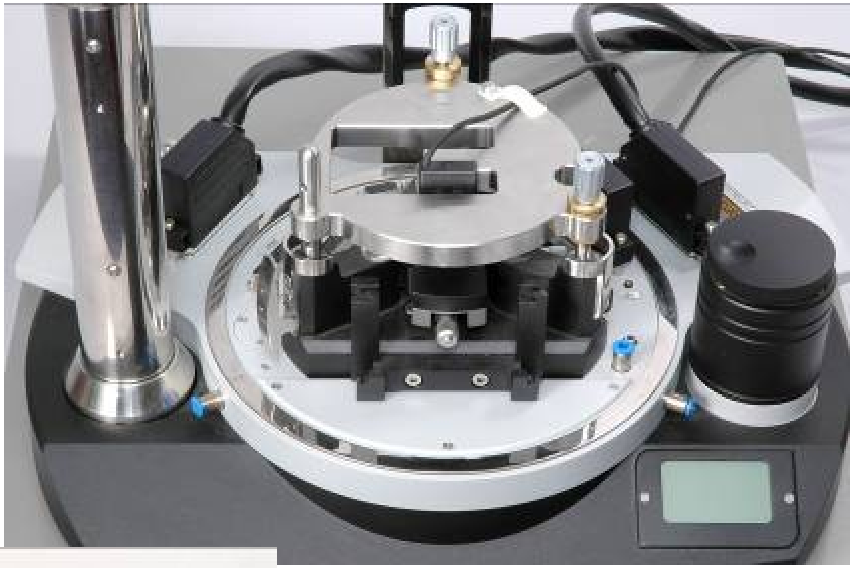


3 µm Scan of a glass fiber/epoxy composite.
topography (left), thermal (right)

20th reason why:- Integrated with Nanoindentors



NTEGRA with nanoindentation head

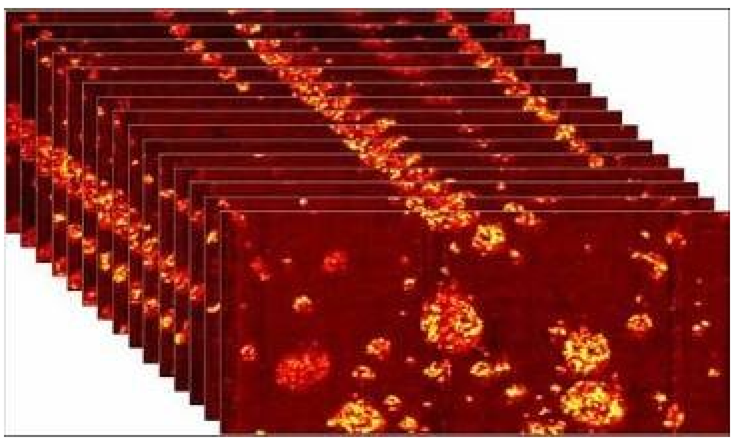


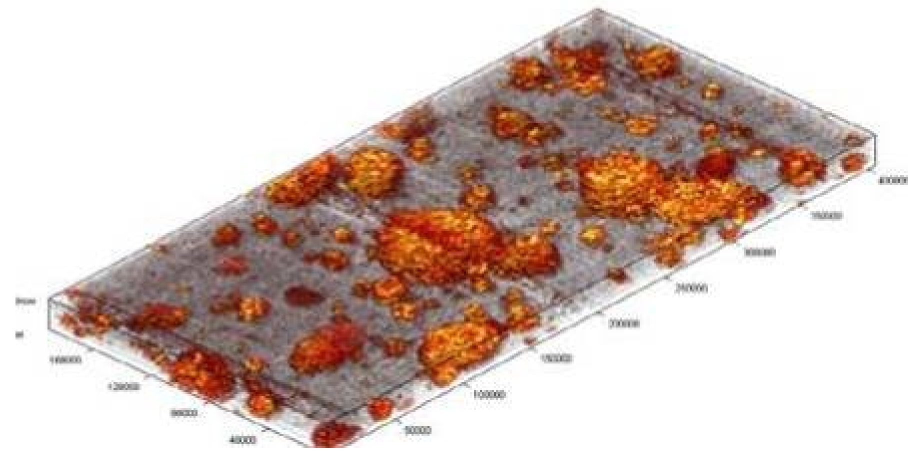
NTEGRA with TriboScope, Hysitron nanoindenter

21th reason why:- Integrated with Microtome

3-D reconstruction:

(Can be animated in actual software application)





PS/HIPS blend with silica

15 sequential AFM images

- Each section is 40x20 µm
- Space between sections: 200 nm

THANK YOU!

NT-MDT

Molecular Devices and Tools for NanoTechnology

YOUR CHOICE IS GRANTED!

E-mail: spm@ntmdt.ru